

Surface Plasmon Resonance Allows to Correlate Molecular Properties With Diffusion Coefficients of Linear Chain Alcohols

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Every biological and physicochemical process occurring in a fluid phase depends on the diffusion coefficient (D) of the species in solution. In the present work, a model to describe and fit the behaviour of D as a function of structure and extensive thermodynamics parameters in binary solutions of linear chain organic molecules is developed. Supporting experimental and computational evidences for this model are obtained by measuring D for a series of n -alcohols through a novel surface plasmon resonance method and molecular dynamics simulations. This allows to propose a kind of combined analysis to explain the dependence of D on various

thermodynamic and structural parameters. The results suggest that for small linear systems in the range from 0 to 200 g mol⁻¹ and under the assumption that the diffusive activation energy is a linear function of mass, D is strictly dependent on the molecular shape and on the relative strength of the solute-solvent intermolecular forces represented by a parameter named R . The newly proposed approach can be utilized to characterize and monitor progressive changes in physicochemical properties for any investigated species upon increasing the dimension of the aggregate/molecule along a certain direction.

Introduction

The dynamics and physicochemical properties of any species in solution are strictly dependent on their chemical potential, which varies as a function of space and time due to mass transport phenomena. Diffusion, convection and advection (in the absence of other external fields that alter the dynamics of a system) are the most relevant mass transport phenomena in many different applications: they play a crucial role in molecular communication,^[1] biological receptor host-guest dynamics and bio-molecular reaction kinetics.^[2]

For this reason, correlating molecular conformational dynamics of molecules with diffusive processes represents a viable and useful way to unravel their microscopic properties. Given a

certain chemical species, it is known that, in absence of convection and advection, the main contribute to the concentration profile evolution over time is diffusion.^[3] Upon any change in molecular shape, a modification of the diffusion coefficient (D) should occur, meaning that this parameter may be used to study progressive changes in solution for aggregation phenomena^[4] and in general for any kind of process in which the three-dimensional structure of a species varies.

This means that as long as the systems under study varies in shape and size, an experimental determination of D allows an indirect determination of both structural and chemical information on homologous series of molecules.

In this perspective, very recently, we have developed a new Surface Plasmon Resonance (SPR) based approach to measure D , named D -SPR hereafter, with high sensitivity and precision (section S1 of the Supporting Information). D -SPR is a suitable technique to assess the conformational and/or oligomerization changes associated with protein unfolding and refolding,^[4] which are commonly associated with the development of various pathological states.^[5,6] D -SPR is also capable of exploring both the diffusive behaviour of folded and unfolded macromolecules and of small solutes, even solvated metal ions.^[7]

Given these premises, in this work we propose a new model to explain the scaling of experimentally measured D as a function of molecular mass and solute-solvent interactions in small to medium sized linear systems. In particular, we tested our model on a series of linear chain alcohols ranging from methanol to n -octanol (any longer linear alcohol chain is almost insoluble in water), demonstrating its validity in the range between 32 and 200 g mol⁻¹ at 296 K. As opposed to the majority of the approaches described in the literature, which are based on various modifications of the classical Stokes–

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Einstein relation, our fitting model is based on the theory of absolute rates.^[8] This means that not only it does not rely on the assumption of rigid spheres for solute and solvent, but also it can be refined as needed to account for effects such as intermolecular interactions and molecular shape. To the best of the authors knowledge, a similar systematic study of structure-activity relations of D for a family of n -alcohols has not been conducted in the literature using SPR methods.^[9] Both the model and the D -SPR method could be applied to any other molecule, given the adequate experimental precautions on the sensor surface functionalization and on the sample preparation. Indeed, as a mean to confirm the validity of our approach, besides the D values obtained from our experiments for alcohols in water, we have also validated our model on the series of linear chain carboxylic acids, retrieving the D data from the literature.

Finally, we coupled the experimental D -SPR data with a computational investigation of the conformational dynamics of the investigated molecules through Molecular Dynamics (MD) simulations, obtaining a novel scaling law for D as a function of configurational entropy (S_c), molecular mass (MM), and pure solute normal boiling point (T_b). Comparing our results with the usual approximation of molecules as spheres of fixed radius in a continuous medium,^[8] a non-trivial dependence of D on molar mass exists for small linear molecules in the molecular weight regime between 0 and 200 g mol^{-1} . By considering these findings, coupled with accurate thermodynamic parameters retrieved from NIST datasets, we propose D -SPR as an experimentally valid method to determine the molecular shape of molecules in aqueous solutions.

Results and Discussion

The progressive increment of methylene units to the carbon backbone modifies the conformational dynamics of any linear organic molecule in water, thus affecting the D value. Moreover, there is a significant dependence of D on concentration in binary solutions of short chain alcohols, up to 1-butanol and, in the present work, the concentration range covered by the

samples spans from 10^{-3} to 10^{-4} in molar fraction, effectively meaning that we are measuring D at almost infinite dilution.

In Figure 1 a simple scheme of the rationale behind the D -SPR experiments is depicted^[7] (see also Section S1 of the Supporting Information).

It is important to highlight that such newly proposed method allows to measure D values with high precision and reproducibility without any limitation, so that it becomes feasible to build and experimentally test a theoretical model developed from the Eyring theory of diffusion in liquids,^[11] to fit the experimental data, obtaining additional unprecedented insight into the thermodynamic features at the base of the diffusion processes.

Results

A complete explanation of the phenomena that determine the scaling of D with molecular mass and shape is not known. In the present work we tackle the problem of evaluating and rationalizing the behaviour of D as a function of various thermodynamics and structural parameters for small linear solutes in nearly ideal water solutions.

Although many studies focus on the interplay between molecular mass, molecular shape and D ,^[12,13] usually they are mostly based either upon power-law fitting methods or on various generalizations of the Stokes–Einstein theory of molecular diffusion (see Section S2 of the Supporting Information).

The aforementioned works build on the well-known Stokes–Einstein equation (Equation (1)) to rationalize and interpret the results of DOSY experiments and, in general, of experimentally determined D values on a great number of test-systems in the low to medium molecular weight regime.

$$D = \frac{k_B T}{6\pi\eta r_h} \quad (1)$$

Where k_B and T are the Boltzmann constant and the absolute temperature, η is the dynamic viscosity of the solvent, and r_h is the hydrodynamic radius of the solute. However, Eq. (1) fails to describe D when the assumptions of continuous

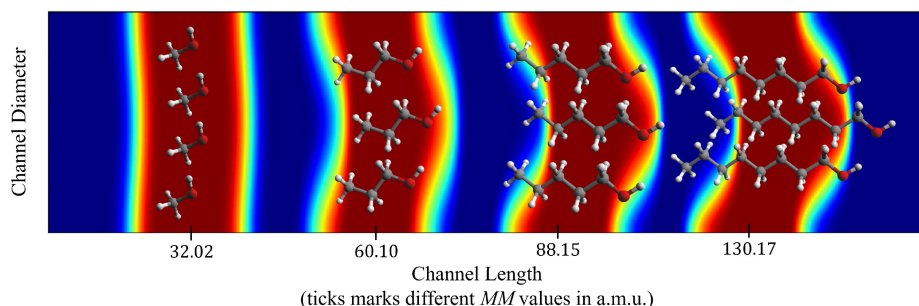


Figure 1. Schematic representation of how the diffusion of differently long alcohols occurs inside the tubing system during a D -SPR experiment. On the x axis each tick represents the molecular weight (in a.m.u.) of the solute. Analyte plugs are color-coded for concentration, with red indicating the highest concentration and blue representing zero concentration. The vanishing wave is excluded for clarity. Each simulation, conducted for this image, utilized finite differences computational schemes with a time step of 8000 ms for 100 steps.^[10] The sole variation among simulations lies in the D value employed to define the governing equation.

structure for the solvent and of rigid spherical shape for the solute are not fulfilled.^[14–16]

In the following, we present one possible fitting model which is based on analyzing D plotted against the eccentricity of the investigated species, along with a fitting model completely based on Eyring's theory of absolute rates. This proposed model is virtually independent on the molecular size of the investigated species, it is grounded in the statistical mechanics theory of Absolute Rates formulated by Eyring and therefore likely generalizable to take into account any possible degree of freedom and, in principle, to consider aggregation phenomena.

We aim to build a model directly from the first principles of the theory of absolute rate processes of Eyring^[8] mainly for two reasons:

- Firstly, given the efforts to make precise predictions based on Molecular Dynamics and statistical mechanics for the physicochemical properties of condensed matter,^[17–19] a model correlating molecular mass and diffusive behaviour based on statistical mechanics is desirable to pave the way for accurate prediction and interpretation of the single structural parameters influencing diffusion.
- Secondly, Eyring's theory is remarkably effective in separating the contributions of different degrees of freedom for a certain solute in solution and is, in principle, capable of mechanistically describing the diffusive process in a concise and clear manner.

Although this work does not aim to derive a complete and rigorous theoretical model to evaluate D from statistical mechanics, we propose a new model to initiate this line of research with the ultimate goal of complementing and supporting results from the experimental D -SPR method and any other technique precise enough to measure D with similar experimental uncertainties.

As a final introductory remark, Bosse and collaborators employed a similar line of reasoning based on the theory of absolute rates to construct a predictive model. They proposed that the activation energy for the process under study varies linearly with mole fraction only in the case of diluted solutions, akin to the approach presented here.^[20] However, while their work primarily focuses on the predictive aspect of the theory and its relation to activity coefficients in solutions far from ideal conditions, the present study relies on the infinite dilution limit to derive a new model correlating D with molecular weight and other related molecular properties.

The main assumptions, better described in Section S3 of the Supporting Information, are as follows:

- The general assumptions of the theory of absolute rates hold.
- The activation energy for diffusion is a linear function of molecular mass.
- The ratio of rotational partition functions in the activated and ground states is different from 1.

From these assumptions and following the derivation in Section S3 of the Supporting Information, we get:

$$D = C\Lambda_{\text{rot}}\left(\frac{1}{x} + \frac{1}{M_w}\right)^{1/2} \exp(-\beta'x) = \quad (2)$$

$$= C(1 + R e^{\beta'x})\left(\frac{1}{x} + \frac{1}{M_w}\right)^{1/2} \exp(-\beta'x) \quad (3)$$

C is a fitting parameter containing information on both the free volume and the average distance travelled by the diffusing molecule, M_w is the solvent molecular weight and β' is a modified thermodynamic beta. Furthermore, Λ_{rot} represents a correction introduced to take into account the ratio of rotational partition functions between activated and ground states and R is a phenomenological parameter, representing how strongly the rotational freedom of linear alcohols is affected by the diffusive processes.^[8] In Eq. (3) we can denote with x any parameter directly proportional with respect to molar mass albeit considering a rescaling of the terms by the appropriate constants.

One significant issue with Eq. (3) is that Λ_{rot} takes negative values for excessively large values of x and negative R and this fact suggests that a different scaling law holds in these conditions. This, however, should not be surprising given the fact that D can only decrease up to the point where diffusive phenomena stops and either phase separation takes place or a viscoelastic behaviour arises. Furthermore, for each case scrutinized by our group, R takes positive values, meaning that Λ_{rot} should be a valid scaling for the investigated systems for any arbitrarily large value of x .

As already stated, our proposed model is found to be appropriate to fit D plotted against any thermodynamic variable linearly scaling with molecular mass (see Figure 2), but perhaps the most interesting plot is done taking D against the pure n-alcohols boiling point (T_b) at 1 atm (Figure 3).

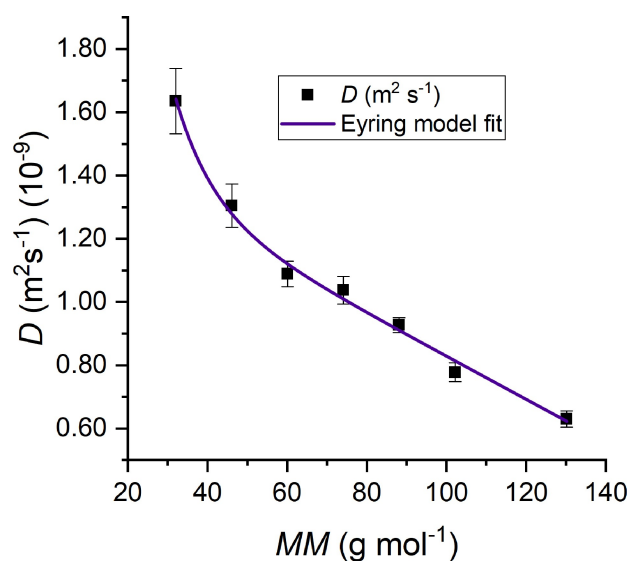


Figure 2. Scatter plot and fit of D against molecular weight for the increasingly longer n-alcohols. The nonlinear fit has been carried out by means of Eq. (3). There is a sensible change in slope for the data trend upon passing from n-propanol to n-butanol.

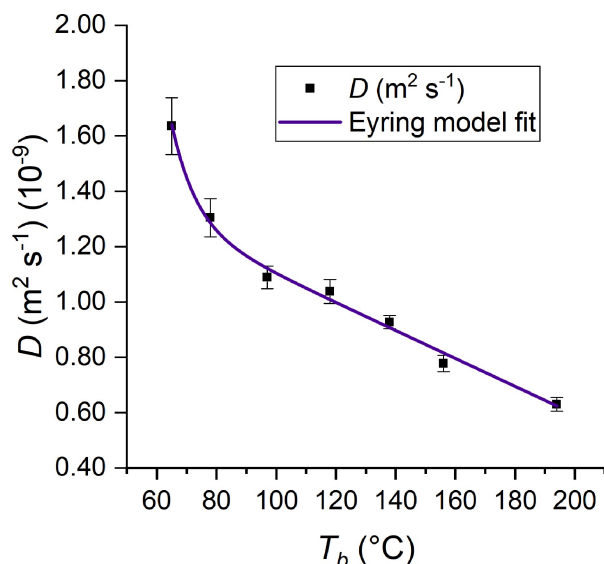


Figure 3. Scatter plot and Fit of D as a function of T_b . The nonlinear fit has been carried out by means of Eq. (3). Again, a significant change in slope upon passing from n-propanol to n-butanol is observed.

One should however notice that the boiling point of a given family of pure substances is not always linear with respect to molar mass, and one should pay close attention in establishing if this is true beforehand. However, even with this limitation, it is fairly common to find tabulated values for the boiling points of pure substances, and even for newly synthesized compounds simple calorimetric techniques may allow researchers to obtain such a data.^[21] A number of ways to estimate β' are possible, based upon the original paper of Steam, Eyring and collaborators,^[11] but a further treatment of those methods is beyond the scope of this work.

Given the form of Eq. (3), the problem of theoretically derive D is then translated to the calculation from first principles of the parameter R , which is certainly of great interest. This parameter conveys information both on the relation between structure and rotational hindrance and on the intermolecular forces acting on the transition state for diffusion, and it is thus of fundamental importance both from the experimental and theoretical perspectives.

Estimation of the R Rotational Parameter

As previously mentioned, in Eq. (3), R is likely influenced by both the chemistry and the structure of the diffusing species in a given solvent. For organic solvents, R is expected to be small, allowing Λ_{rot} to approximate the ideal case. However, in structured solvents, this may not hold true due to the network of intermolecular interactions that “cage” the solute, potentially allowing more rotational freedom in the activated diffusion state. To explore this hypothesis, we compared different R

values across various D for different series of solutes. We utilized the following datasets:

1. Our experimental set of D values for n-alcohols in water, including the self-diffusion coefficient of water.
2. A set of D values for linear carboxylic acids in water.^[22,23]
3. A set of D values for n-alcohols in cyclohexane, obtained from the literature.^[24]

We included the self-diffusion coefficient of water in the n-alcohol series to study how R varies in a series of molecules that can be considered as water molecules with one hydrogen replaced by an alkyl chain of increasing length. Notably, the R value obtained by fitting with and without the water data point is similar, considering the error in parametric estimation. Including water, R is estimated to be 0.34 ± 0.06 , while excluding water, the value is 0.2 ± 0.1 .

For the series of linear carboxylic acids in water, we used the D values reported by Bidstrup et al.^[22] The final data point, obtained from another study, corresponds to the D value for n-heptandioic acid.^[23] Although this molecule is not strictly a monocarboxylic acid, the spacing between the terminal groups is such that electrostatic interactions should not significantly affect the conformational freedom of the linear carbon chain.

To estimate R we employed Eq. (3) to fit each data set plotted against MM and the results of these fitting procedures are presented in Figure 4.

The results in Table 1 clearly show that R decreases when moving from alcohols to carboxylic acids in water, indicating that factors beyond conformational freedom influence the diffusive behaviour of these molecular classes. Furthermore, as hypothesized, R significantly depends on the solvent's nature, even for the same series of molecules. For n-alcohols in cyclohexane, R is approximately half of its value in water, suggesting that, at equal molecular weights, the influence of Λ_{rot} in Eq. (3) is less significant.

A clear understanding of R is crucial for comprehending the diffusive properties of molecules in solution, with intermolecular interactions playing a significant role in determining R variations.

Molecular Shape and D-SPR Experiments

In order to tackle the problem of diffusion in binary systems by means of Eq. (3), it is mandatory to employ a reliable experimental method to determine D for a series of molecules with similar chemistries. The model systems employed here are the n-alcohols from methanol up to 1-octanol, and the experimental technique chosen is the D -SPR method developed by Zingale et al.^[7] We refer readers to the Supporting Information (Section S1) for further experimental details, a

Table 1. Summary of fitted R values, each extrapolated using Eq. (3) with appropriate temperature and M_w values.

D^{solvent} species	D^{water} $R\text{-OH}$	D^{water} $R\text{-COOH}$	$D^{\text{cyclohexane}}$ $R\text{-OH}$
R	0.34 ± 0.06	0.25 ± 0.03	0.15 ± 0.01

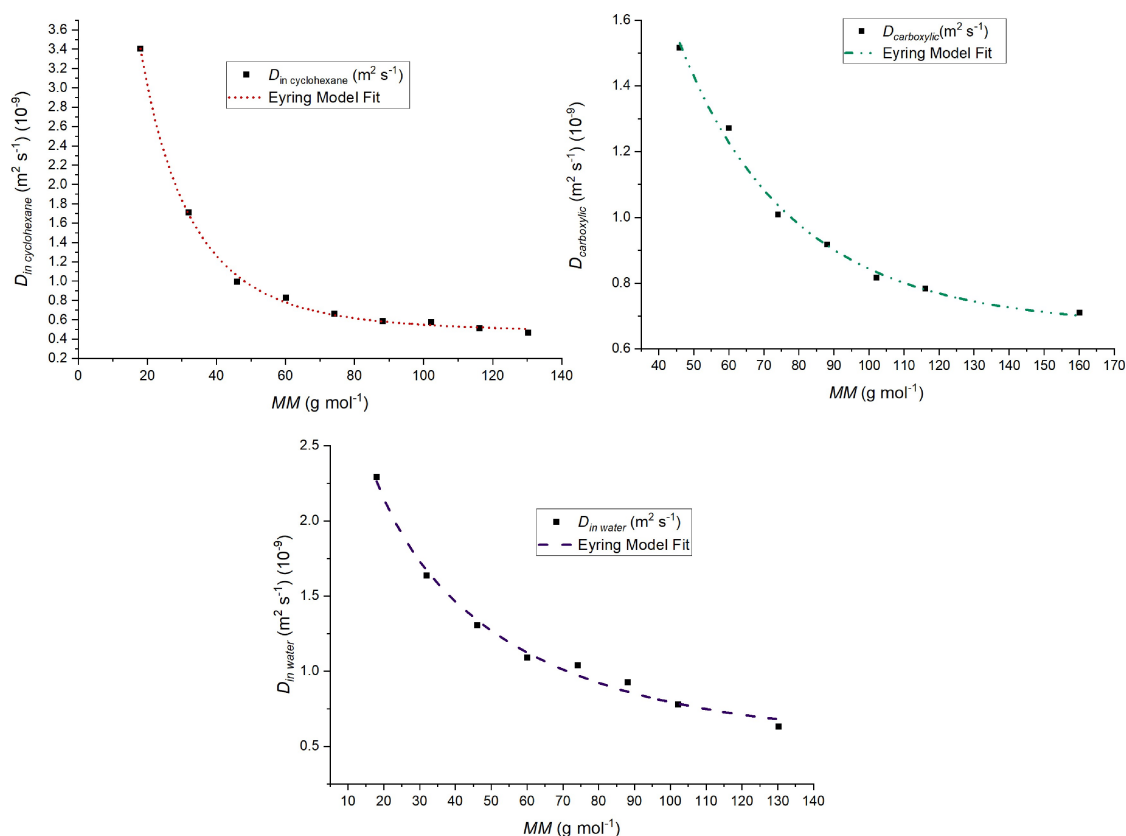


Figure 4. From top left to bottom center: scatter plots of D vs MM for n-alcohols in cyclohexane, linear carboxylic acids in water, n-alcohols in water.

comparison of its advantages and a brief explanation of the working principle for this technique.

Although several studies have been investigating the diffusion properties of various alcohols,^[9,25–29] to the best of the authors knowledge this is the first time that the binary D values for a series of alcohols with increasing chain length in water is estimated in a consistent and reproducible way through SPR methods.

To assess the reliability of our data, we performed a comparison with D values in the literature (Section S4 of the Supporting Information). Given the plausible reliability of the acquired data, as already mentioned, we performed the fitting of D as a function of various thermodynamics parameters linearly scaling with molar mass by means of Eq. (3), finding that our proposed model fits very well the data against both MM and T_b as evident from Figure 2 and 3.

As evident from the previous considerations, given a linear increase in molecular mass, the D values of n-alcohols in water behave in manner that deviates from the simple Stokes–Einstein relation and most of its modified versions. In order to understand why this happens, we tried to complement the previous model with the aid of classical Molecular Dynamics (MD) simulations. For each of the systems aside 1-octanol, we performed MD simulations with the GROMACS package (see Section S5 of the Supporting Information), acquiring a total of six trajectories, from methanol up to 1-hexanol. These trajectories were then analyzed to extrapolate D values, S_c values (by

means of the Quasi-Harmonic Approximation in Cartesian coordinates), and average end to end distances. A very good agreement in the general trend is observed for both simulated and experimental D values, albeit the absolute values of the coefficients are not in perfect agreement (see Table 2 and Figure 5). This is however expected (Section S5 of the Supporting Information) due to the simple assumptions made in the trajectory analysis to determine D , namely, considering just the

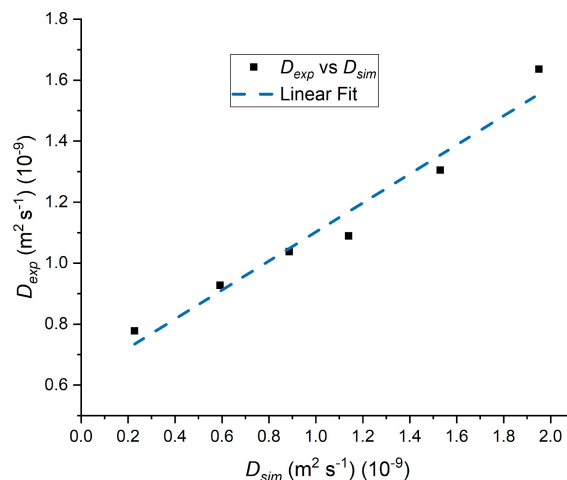


Figure 5. Correlation plot of D for the experimental and MD simulated values. A linear trend with a slope of ≈ 0.5 is observed.

Table 2. Resuming Table for the experimental and simulated D values at 296 K.

$N^{\circ}C$	MM $g\ mol^{-1}$	D_{exp} $m^2\ s^{-1}$	$\sigma_{D_{exp}}$ $m^2\ s^{-1}$	D_{sim} $m^2\ s^{-1}$	$\sigma_{D_{sim}}$ $m^2\ s^{-1}$
1	32.04	$1.60 \cdot 10^{-9}$	$0.10 \cdot 10^{-9}$	$1.95 \cdot 10^{-9}$	$8.39 \cdot 10^{-11}$
2	46.07	$1.30 \cdot 10^{-9}$	$0.07 \cdot 10^{-9}$	$1.53 \cdot 10^{-9}$	$1.59 \cdot 10^{-10}$
3	60.1	$1.09 \cdot 10^{-9}$	$0.04 \cdot 10^{-9}$	$1.14 \cdot 10^{-9}$	$7.29 \cdot 10^{-11}$
4	74.12	$1.04 \cdot 10^{-9}$	$0.04 \cdot 10^{-9}$	$8.87 \cdot 10^{-10}$	$5.99 \cdot 10^{-11}$
5	88.15	$0.93 \cdot 10^{-9}$	$0.02 \cdot 10^{-9}$	$5.93 \cdot 10^{-10}$	$6.05 \cdot 10^{-11}$
6	102.18	$0.78 \cdot 10^{-9}$	$0.03 \cdot 10^{-9}$	$2.28 \cdot 10^{-10}$	$1.71 \cdot 10^{-11}$
8	130.23	$0.63 \cdot 10^{-9}$	$0.02 \cdot 10^{-9}$		

translational degrees of freedom of the centers of mass for each molecule in evaluating the MSD (Mean Square Displacement).

As evident from Table 1, the order of magnitude for both computational and experimental values is very similar, even if the first entries for the simulated coefficients overestimate D_{exp} while the last entries underestimate it.

On the other hand, examining the plots of D versus the various structural and thermodynamics parameters previously cited, it is clear that upon passing from 1-butanol to the subsequent series of alcohols, whatever the data trend is, there is a sensible change in its slope. This is further verified by means of a log-log plot analysis of $\log(D)$ against $\log(V)$ where V is the molar volume of each alcohol, obtained from the experimental densities in the literature.^[30] For a perfect Stokes–Einstein equation, by linear fit of a Log–Log plot, a single slope of $-1/3$ should be obtained. In our case, two distinct slopes of ≈ -0.54 and ≈ -0.94 were obtained for different intervals of our dataset, as reported in Figure 6.

It is common, in the literature, to assume that for large enough globular molecules, one might approximate the molecular shape to that of a perfect sphere moving in a continuous solvent, and from these considerations stem the already mentioned Eq. (1) as well as the Evans model (see also Section S2 of the Supporting Information).

By employing the latter, however, the accuracy of estimation for D is as high as 20% with respect to the experimental

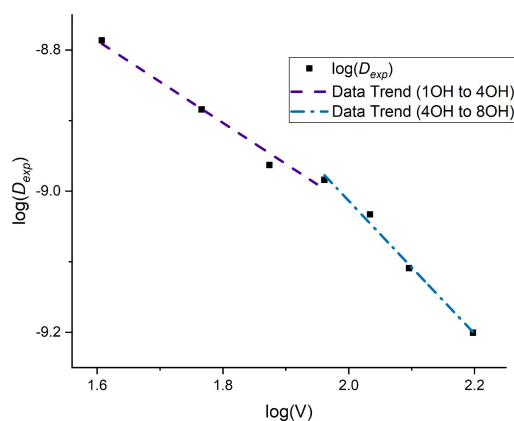


Figure 6. Scatter plot of $\log(D)$ vs. $\log(V)$. In violet: linear fit from methanol up to 1-butanol. In blue: linear fit from 1-butanol up to 1-octanol.

values for each alcohol, as evident in Figure 7. This suggests that, as already stated by Evans et. al, there is a combination of factors that determine the actual scaling for D as a function of molecular weight for different classes of molecules.

The same functional trend is somewhat found in all of the plots of D as a function of different thermodynamics parameters (Figure 8). This is not in accordance with the expected trend for $D(S_c)$ due to the Adam–Gibbs relation.^[31] The dependence of S_c on the molecular mass has been established by means of the QH approximation and it has been proven to be of a linear kind (see Section S5 of the Supporting Information).

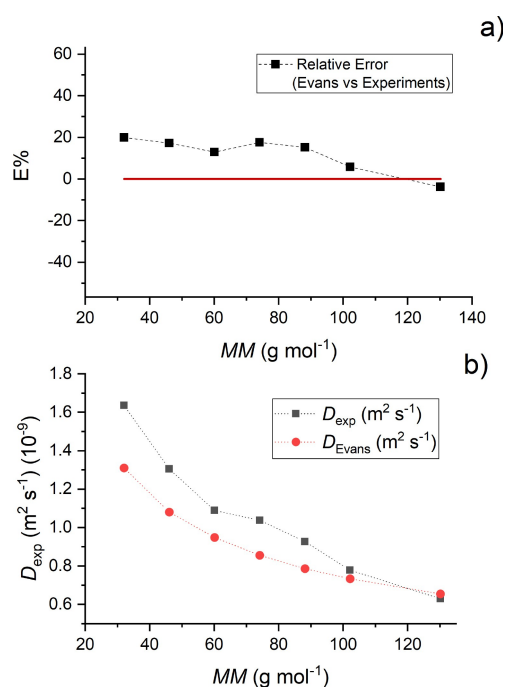


Figure 7. a) Scatter Plot of the accuracy of estimation for D as predicted by the Evans model with respect to our experimentally determined data points. b) Scatter plot of D vs MM for n -alcohols as predicted by Evans et al. (in red) and as measured by means of the D -SPR technique.

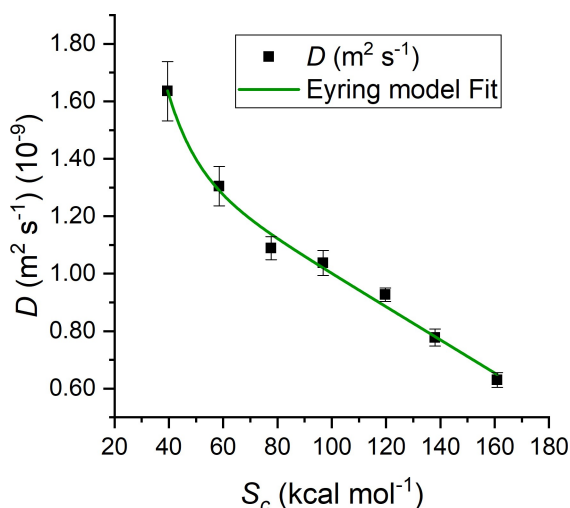


Figure 8. Scatter plot and fit of experimental D values against configurational entropy S_c . The nonlinear fit has been carried out by means of Eq. (3). S_c for 1-octanol has been extrapolated from the linear trend of $S_c(MM)$ (see Section S5 of the Supporting Information).

Discussion

Rationally speaking and given our premises, one could not expect Eq. (1) to be accurate, so to further investigate the behaviour of D as a function of molecular shape, we assumed that each alcohol other than methanol is approximable to a prolate spheroid of minor semi-axis $a = r_{exp}$ and major semi-axis b equal to the average semi-distance between the one methyl hydrogen of the chain and the hydroxyl hydrogen (see Section S6 of the Supporting Information and Figure 9).

One of the many geometrical descriptors for quadrics in the Euclidean space is the so called eccentricity (e'). Given a generic quadric, the eccentricity directly describes how much does it differ from a perfect sphere. For ellipsoids, and in particular for prolate spheroids, $0 \leq e' \leq 1$ and the actual formula is:

$$e' = \sqrt{1 - \frac{a^2}{b^2}}, \quad b \geq a \quad (4)$$

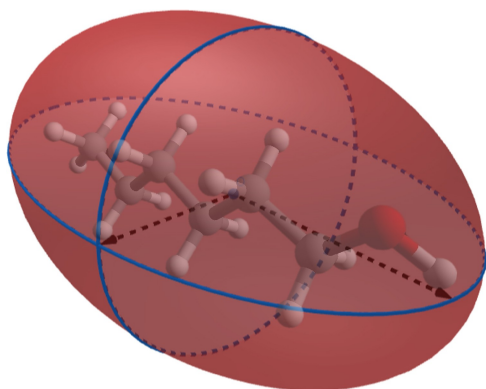


Figure 9. Graphical depiction of a prolate spheroid, the shortest dashed vector is the minor semi-axis a , while the longest dashed vector represents the major semi-axis b .

Evaluating e' for each alcohol and plotting D as a function of eccentricity (Figure 10), one immediately notices that a steep decrease in D is observable as the molecular shape tends to the limit of unitary eccentricity, thus effectively becoming large molecules (governed by fluid-dynamics phenomena).

One might speculate just from intuitive geometry that the eccentricity varies strongly upon increasing the chain length from 1 to 3 carbon atoms, and this is indeed the case as observable in Figure 10. To see how well the prolate spheroid approximation fits with respect to the experimental trend, we employed a shape corrected version of Eq. (1) (see Section S5 of the Supporting Information):

$$D_{prolate} = \frac{k_B T}{6\pi\eta} \frac{(a/b)^{2/3}}{\sqrt{a^2 b}} \ln \frac{1 + e'}{e'} \quad (5)$$

From Figure 11 it is evident that even if the experimental trend is somewhat similar to the theoretical one, the latter overestimates D by a factor as high as 1.3, this is however quite reasonable if we take into account that we neglected intermolecular interactions and the local structure of the solvent.

An overestimation of D is not consistent with the original results of Eyring for aqueous solutions, based upon a semi-empirical equation derived in the original work.^[11]

It is worth noticing, though, that even within the more sophisticated framework of Evans model, the result is slightly underestimated, meaning that taking into account the finite size of the solvent, the results are qualitatively similar to those of Eyring.

In his original work on diffusive phenomena as activated processes, Eyring states that upon passing from the ground state to the activated “transition state” for the diffusive process, the rotational freedom of each solute molecule probably increases as an appropriate number of water-water hydrogen-bonds breaks to allow for the solute to diffuse. A similar argument has been employed here to rationalize and comprehend the behaviour of $D(x)$.

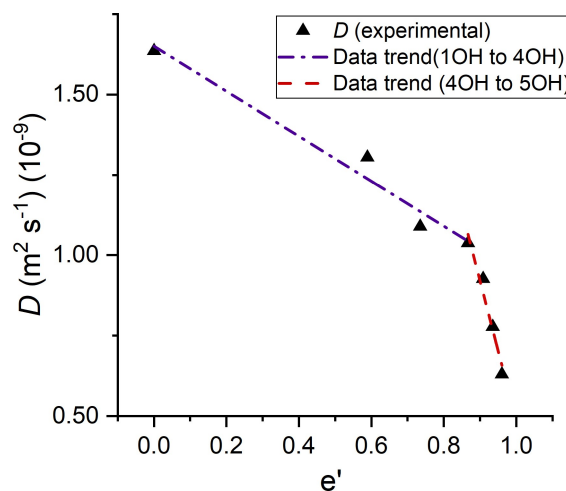


Figure 10. Scatter plot of D against eccentricity with particular emphasis on the different data trends and slopes.

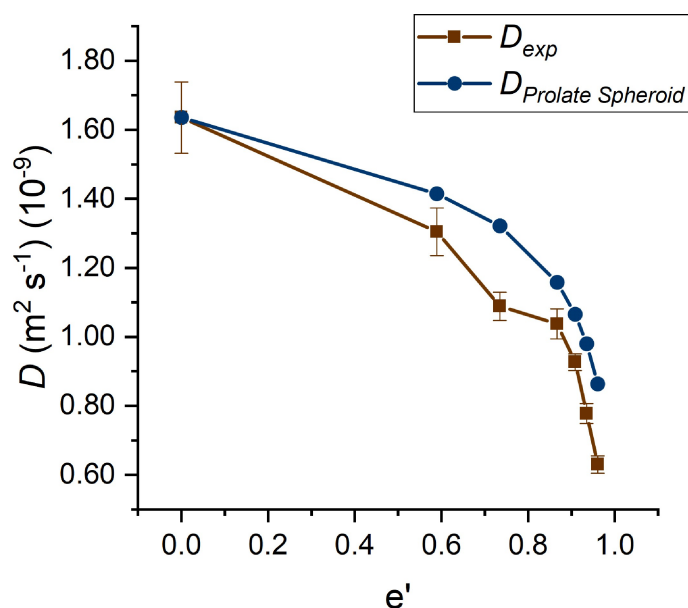


Figure 11. Scatter plot of D against eccentricity (e') for the experimental D -SPR (squares) data and for the prolate spheroid approximation results (circles). In blue, the D values predicted by means of Eq. (S40) (see Section S6 of the supporting material), in red, the experimental D values.

These considerations, however, suggest to us that even in its various modified versions, the Stokes–Einstein relation alone is not enough to rationalize the behaviour of small solutes in structured solvents, meaning that a more sophisticated statistical mechanics treatment should be employed to accurately depict D as a function of molecular shape and intermolecular forces.

Given that a complete explanation of the relationship between shape and diffusion is not yet achievable, one applicable approach is to study the functional form of $D(x)$, as done by using Eq. (3), with x taken as a suitable extensive molecular property. If we can measure D using the D -SPR technique (or any other precise method) for any species under study (molecules, macromolecules, or supramolecular aggregates), studying the functional form of $D(x)$ provides information on R , β' , and $\mathcal{P}$, each bearing structural and chemical information about the investigated species.

Furthermore, by coupling experimentally determined D values with computationally derived structural parameters such as eccentricity, a complementary approach to characterize the interplay between shape and diffusion becomes feasible, as here demonstrated in the case of linear chain n -alcohols. Geometrical descriptors from simulations are relatively easy to compute, while D values are somewhat harder to derive accurately. Once the descriptor is determined, plotting experimentally determined D versus the descriptor can help understand how D changes as a function of geometrical parameters, providing direct insight into the dependence of diffusive behaviour on a more realistic depiction of molecular shape. This kind of study is suitable for unveiling the interplay between molecular structure and intermolecular forces in increasingly longer soluble non-branched systems such as linear synthetic

oligomers, peptides, intrinsically disordered proteins (IDPs), and supramolecular aggregates.

The key idea for future applications of this analysis is to conduct carefully designed full-atomistic MD simulations and use them to predict a plausible value for R , along with a few suitable sets of geometric descriptors for a preliminary approximation of molecular shape (e.g., eccentricity or aspect ratio for cylinders and spheroids).

With these speculations in mind, D -SPR experiments can either confirm or refute the simulated results by fitting through Eq. (3), providing a clear and simple way to discern whether a simulation result is correct or based on biased assumptions. Similar arguments and analyses should and could be applied to differently shaped structures in the future.

Conclusions

A novel kind of structural and thermodynamic analysis has been developed based on a newly derived model (Eq. (3)). The strength of this work lies in the relative ease and repeatability of determining experimental D values, facilitated by the newly proposed D -SPR approach.

Experimental data and molecular simulations suggest that a non-trivial dependence on molar mass exists for small linear molecules in the regime between 0 and 200 $g\ mol^{-1}$. Notably, this behaviour is not the one predicted by means of the Stokes–Einstein equation.

With the aid of MD, a large number of more sophisticated analyses are possible, allowing the estimation of molecular shape, under the constraint of a plausible guess on the general form of such shape. This observation suggests that, under the assumption of sufficiently fast relaxation times for the investigated systems, diffusion may be used to characterize molecular shape change in increasingly larger systems. In the future, analyses of complex systems such as proteins and polysaccharides, along with the analogous series of linear monofunctional organic molecules (n -thiols, n -methyl-esters) will be conducted, facilitating a better understanding of the relationship between molecular size, conformation, diffusion, and aggregation.

Experimental Section

For the sake of clarity and conciseness, all the technical details for the Experimental procedures have been moved to the Supporting Information file (Sections S1 and S4).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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